

LAB ASSIGNMENT 8

PROBLEM 1: Create State of art table for all the algorithms used in the lab (Classification Algorithm).

CODE:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score, precision_score,
recall_score, f1_score

df_d = pd.read_csv('diabetes_dataset.csv')
print("Diabetes")
print(df_d.head())
print("Missing values")
print(df_d.isnull().sum())
df_d.fillna(df_d.mean(numeric_only=True), inplace=True)
df_s = pd.read_csv('Social_Network_Ads.csv')
print("\nSocial Network Ads")
print(df_s.head())
print("Missing values")
print(df_s.isnull().sum())
df_s.fillna(df_s.mean(numeric_only=True), inplace=True)
df_t = pd.read_csv('titanic.csv')
print("\nTitanic")
print(df_t.head())
print("Missing values")
print(df_t.isnull().sum())
df_t['Age'] = df_t['Age'].fillna(df_t['Age'].median())
df_t['Embarked'] =
df_t['Embarked'].fillna(df_t['Embarked'].mode()[0])
df_t = df_t.drop(columns=['Cabin'])
print("Missing values (after analysis)")
```

```
print(df_t.isnull().sum())

datasets = {
    "Diabetes Dataset": df_d,
    "Social Dataset": df_s,
    "Titanic Dataset": df_t
}

models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "Naive Bayes": GaussianNB(),
    "SVM": SVC(kernel='rbf')
}

results = []
for dataset_name, df in datasets.items():
    for col in df.columns:
        if df[col].dtype == 'object':
            df[col] = LabelEncoder().fit_transform(df[col])
    X = df.iloc[:, :-1]
    y = df.iloc[:, -1]

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X_train)
    X_test = scaler.transform(X_test)
    for model_name, model in models.items():
        model.fit(X_train, y_train)
        y_pred = model.predict(X_test)
        acc = accuracy_score(y_test, y_pred) * 100
        prec = precision_score(y_test, y_pred, average='weighted',
zero_division=0)
        rec = recall_score(y_test, y_pred, average='weighted',
zero_division=0)
        f1 = f1_score(y_test, y_pred, average='weighted',
zero_division=0)

        results.append([
            model_name,
            dataset_name,
            round(acc, 2),
```

```
        round(prec, 2),
        round(rec, 2),
        round(f1, 2)
    ])
final_df = pd.DataFrame(results, columns=[
    "Model / Method",
    "Dataset",
    "Accuracy (%)",
    "Precision",
    "Recall",
    "F1 Score"
])
```

OUTPUT:

```
PS D:\WITJ ACADEMICS\OPENLEARN> python -u "d:\WITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_9_1.py"
```

```
Diabetes
Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  BMI  DiabetesPedigreeFunction  Age  Outcome
0            6    148             72             35         0    33.6                0.627    50      1
1            1     85             66             29         0    26.6                0.351    31      0
2            8    183             64              0         0    23.3                0.672    32      1
3            1     89             66             23         94    28.1                0.167    21      0
4            0    137             40             35        168    43.1                2.288    33      1
```

```
Missing values
```

```
Pregnancies      0
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI               0
DiabetesPedigreeFunction  0
Age               0
Outcome           0
dtype: int64
```

```
Social Network Ads
```

```
User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510  Male   19           19000         0
1  15810944  Male   35           20000         0
2  15668575  Female  26           43000         0
3  15603246  Female  27           57000         0
4  15804002  Male   19           76000         0
```

```
Missing values
```

```
User ID          0
Gender           0
Age              0
EstimatedSalary  0
Purchased        0
dtype: int64
```

```
Titanic
```

```
PassengerId  Survived  Pclass  Name  Sex  Age  SibSp  Parch  Ticket  Fare  Cabin  Embarked
0            1         0        3  Braund, Mr. Owen Harris  male  22.0    1    0      A/5 21171  7.2500  NaN      S
1            2         1        1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1    0      PC 17599  71.2833  C85      C
2            3         1        3  Heikkinen, Miss. Laina  female  26.0    0    0  STON/O2. 3101282  7.9250  NaN      S
3            4         1        1  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0    1    0      113803  53.1000  C123      S
4            5         0        3  Allen, Mr. William Henry  male  35.0    0    0      373450  8.0500  NaN      S
```

```
Missing values
```

```
PassengerId      0
Survived          0
Pclass            0
Name              0
Sex               0
Age              177
SibSp             0
Parch             0
Ticket            0
Fare              0
Cabin            687
Embarked          2
dtype: int64
Missing values (after analysis)
PassengerId      0
Survived          0
Pclass            0
Name              0
Sex               0
Age              0
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          0
dtype: int64
```

TABLE

	Model / Method	Dataset	Accuracy (%)	Precision	Recall	F1 Score
0	Logistic Regression	Diabetes Dataset	75.32	0.76	0.75	0.75
1	KNN	Diabetes Dataset	69.48	0.69	0.69	0.69
2	Naive Bayes	Diabetes Dataset	76.62	0.77	0.77	0.77
3	SVM	Diabetes Dataset	73.38	0.73	0.73	0.73
4	Logistic Regression	Social Dataset	88.75	0.89	0.89	0.88
5	KNN	Social Dataset	92.50	0.92	0.93	0.92
6	Naive Bayes	Social Dataset	91.25	0.91	0.91	0.91
7	SVM	Social Dataset	92.50	0.93	0.93	0.93
8	Logistic Regression	Titanic Dataset	67.60	0.63	0.68	0.56
9	KNN	Titanic Dataset	68.16	0.66	0.68	0.65
10	Naive Bayes	Titanic Dataset	52.51	0.65	0.53	0.55
11	SVM	Titanic Dataset	69.27	0.64	0.69	0.60

